ITAI 3377 – AI at the Edge & IIOT Env.

Prof. Sitaram Ayyagari

Natalia Solorzano W207818526

Spring 2026 – 17181

L07: IIoT Network Analysis: Age of Information and Reliability Trade-offs

1. Key Concept

Age of Information (AoI) measures how *fresh* data is at the receiver. It increases over time when no updates are received and resets when a new update arrives. Unlike delay or throughput, AoI focuses on how current the information is, which is critical in IIoT systems where outdated data can lead to incorrect decisions.

There are two main traffic types:

- **AoI-oriented traffic (TAoI):** periodic updates (e.g., sensors)
- **Deadline-oriented traffic (TD):** urgent messages with strict deadlines (e.g., alarms)

These two compete for network resources, creating a trade-off between:

- **Low AoI (fresh data)**
- **Low Packet Loss Probability (PLP) (reliable delivery)**

2. Key Data Insights

Analysis of ~10,000 simulated network records revealed:

- **Transmission probability (p2)** has the strongest impact:
  - Higher p2 → lower AoI (more frequent updates)
  - But → higher PLP (more collisions)
- **Capture threshold (γ)** affects decoding difficulty:
  - Higher γ → worse AoI and higher PLP
  - Lower γ improves overall performance
- **Trade-off behavior:**
  - Optimizing for **low AoI** increases **packet loss**
  - Optimizing for **low PLP** increases **data staleness**
- **Other factors:**
  - Better **channel quality** improves both AoI and PLP
  - More **nodes** increase interference → higher AoI

3. Machine Learning Results

A **Random Forest model** was used to predict AoI and achieved strong performance ($R^2 \approx 0.85$–0.95).

Most important features:

1. Transmission probability (p2) – strongest factor
2. Channel quality
3. Capture threshold (γ)
4. Number of nodes

A **neural network model** was also tested to predict both AoI and PLP simultaneously. While slightly less accurate for AoI alone, it is more useful for real-time decision-making since it models the trade-off directly.

4. Key Insights

- The **main control variable** is transmission probability (p2)
- **System performance is a balancing act** between freshness (AoI) and reliability (PLP)
- **Network conditions (γ, channel quality, node count)** shape but do not dominate the trade-off

5. Optimization Strategies

- **Adaptive transmission control:**
  Dynamically adjust p2 based on network conditions
  - High p2 when no urgent traffic → better freshness
  - Low p2 when deadlines matter → better reliability
- **Improve receiver capability:**
  Lowering capture threshold (γ) improves both AoI and PLP simultaneously

6. Real-World Applications

- **Smart factories:**
  Balance sensor updates (fresh data) with emergency alerts (reliability)
- **Autonomous robots (AGVs):**
  Maintain up-to-date location data while ensuring critical stop commands are delivered

7. Conclusion

The study shows that IIoT networks must carefully balance **data freshness and reliability**. Transmission probability is the most powerful tuning parameter, while improvements in channel conditions and receiver design can enhance overall system performance. Machine learning models further support real-time optimization of this trade-off.

References

Farag, H., Ali, S. M., & Stefanovic, C. (2023). *On the analysis of AoI-reliability tradeoff in heterogeneous IIoT networks*. **arXiv**. https://arxiv.org/abs/2311.13336

Sisinni, E., Saifullah, A., Han, S., Jennehag, U., & Gidlund, M. (2018). Industrial internet of things: Challenges, opportunities, and directions. *IEEE Transactions on Industrial Informatics, 14*(11), 4724–4734.

Sun, Y., Uysal-Biyikoglu, E., Yates, R. D., Koksal, C. E., & Shroff, N. B. (2017). Update or wait: How to keep your data fresh. *IEEE Transactions on Information Theory, 63*(11), 7492–7508.

Maatouk, A., Kriouile, S., Assaad, M., & Ephremides, A. (2020). The age of incorrect information: A new performance metric for status updates. *IEEE/ACM Transactions on Networking, 28*(5), 2215–2228.

Zorzi, M., & Rao, R. (1994). Capture and retransmission control in mobile radio. *IEEE Journal on Selected Areas in Communications, 12*(8), 1289–1298.

Tse, D., & Viswanath, P. (2005). *Fundamentals of wireless communication*. Cambridge University Press.